Technical Note 8: Powder Metallurgy

Adapted from *Metals Handbook*, Ninth Edition, Volume 7, and reviewed by J.P. Beckman, Crucible Compaction Metals

Titanium products fabricated by powder metallurgy (P/M) techniques can be divided into two general categories:

- Blended elemental P/M, in which a blend of elemental powders, along with master alloy or other desired additions, is cold pressed into shape and subsequently sintered to higher density and uniform chemistry
- Prealloyed P/M in which prealloyed powder produced by various techniques is consolidated by either cold pressing and sintering or HIP. The powder morphology determines the type of compaction method which can be used: Irregularly shaped powders can be cold pressed into green shapes and subsequently sintered. Spherical powders will not compact to green shapes and are typically containerized and consolidated by HIP

To date, most components produced by the various P/M methods have been made from the widely used Ti-6Al-4V structural alloy. However, P/M technology is also very well suited for other alloys, such as the high-strength β alloys and the high-temperature near-α alloys.

P/M technology also offers a promising solution to the poor workability of titanium aluminides by the technique of reactive powder processing (*Journal of Metals*, May 1993, p 52-56). In this process, elemental powders with the desired composition are compacted and worked into a Ti-Al composite form, which is easily machined or reformed into different shapes. Reactive sintering is then conducted to obtain the desired microstructure of TiAl and Ti_3Al intermetallics.

Other applications of P/M processing include mechanical alloying and the production of particulate-reinforced titanium composities such as CermeTi®. These and other developments are reviewed in a recent article by Froes and Suryanarayana (Powder Processing of Titanium Alloys, *Reviews in Particulate Materials*, Vol 1, MPIF, 1993, p 223-276). Full coverage of these techniques is beyond the scope of this Technical Note, which only provides a brief review of P/M technology and some property data for the P/M composite CermeTi®. P/M property data for more conventional Ti alloys such as Ti-6Al-4V are covered in the alloy datasheets. In general, the mechanical properties of titanium P/M products depend on alloy composition, density and final microstructure of the compact. The final density and microstructure depend on the nature of the powder, on the specific consolidation technique employed, and on postconsolidation thermal or mechanical treatments.

Elemental Powder

Elemental Powder Production. Historically, inexpensive titanium elemental powder has been produced from fines generated during the crushing of titanium sponge. Elemental titanium powder made from sponge fines is typically −100 mesh (<150 microns) and has an irregular angular shape (Fig. 1). Sodium reduced sponge yields larger amounts of fines and typically has lower chloride content than does magnesium reduced sponge. However, magnesium reduced sponge produced by the newer vacuum distillation process (VDP) has even lower levels of chloride. Residual chloride content causes porosity in the consolidated P/M parts. This porosity often results in a degradation of the mechanical properties as well as difficulties in welding. However, the quality of P/M parts made from blended elemental powders using sponge fines is acceptable for many applications except those which are fatigue-critical.

Due to plant closings, sponge fines are less readily available and alternative production methods for elemental titanium powders are being evaluated. These include various electrolytic and vapor phase reduction methods. Of course, elemental

Fig. 1 SEM micrograph of sponge fines (-325 mesh RMI-020 commercially pure titanium powder) 500x

titanium powders can also be produced by the same methods used for producing prealloyed powders which are outlined below.

Prealloyed Powder

Each particle of the prealloyed (PA) titanium powder exhibits the composition of the alloy. Commercial production of PA powders involves either comminution of hydrogen embrittled alloy stock known as the Hydride Dehydride process (HDH), or various methods of melting and rapid solidification (RS). The HDH method produces irregularly shaped blocky particles (Fig. 2) which can be pressed and sintered, while the RS methods produce spherical powders (Fig. 3) which cannot.

Hydride Dehydride (HDH)

Comminution of brittle materials can be used to produce powder. Because titanium and its most common alloys are generally ductile, they must be converted to a brittle state prior to comminution. In the HDH process, this is accomplished by hydrogen embrittlement. Titanium (or a titanium alloy) is hydrided by heating in a hydrogen atmosphere. A variety of solids can be used as starting stock, although light-gage stock has greater surface area which speeds the solid state diffusion process. After hydriding, the product is extremely friable and either spontaneously reduces to powder or is easily pulverized. The powder is then dehydrided (degassed) by heating in a vacuum.

HDH powder is similar in morphology to sponge fines. Both have low packing density (20% - 40%) and low flow rates due to their irregular particle morphology but both are easily cold pressed into green shapes. However, depending on the starting stock, HDH powder may or may not be encumbered with residual chloride content and its resultant porosity on consolidation. The HDH process itself does tend to increase the oxygen content, and, depending on starting stock, other contaminants may be present which could make HDH powders unsuitable for critical applications.

Rapid Solidification (RS)

Rapid solidification produces spherical powder (Fig. 3), which flows readily with minimal bridging tendency, and packs to a very consistent density of approximately 65%. Rapid solidification methods can be categorized as either those that use centrifugal atomization, or those that don't. For those methods that do use centrifugal atomization, either the starting stock is rotated (e.g. PREP) or the collection device is rotated (e.g. RSR). On the other hand, gas atomization (GA) does not use centrifugal atomization. A molten metal stream flows through a nozzle into a ring of gas jets where the liquid metal is atomized by gas pressure.

A comprehensive overview of various RS methods is found in "Rapid Solidification Processing of Titanium Alloys," (C. Suryanarayana, F.H. Froes, R.G. Rowe, published in *International Materials Reviews*, 1991, Vol 36 (No. 3). Most of the RS processes for making Ti PA powder which are listed in this review article are not commercial. Commercial RS methods of powder production include PREP at Nuclear Metals Inc. and gas atomization (GA) at Crucible Materials Corporation.

PREP. A variety of centrifugal atomi-

Fig. 2 SEM micrograph of -270 mesh WahChang HDH commercially pure titanium powder 500x

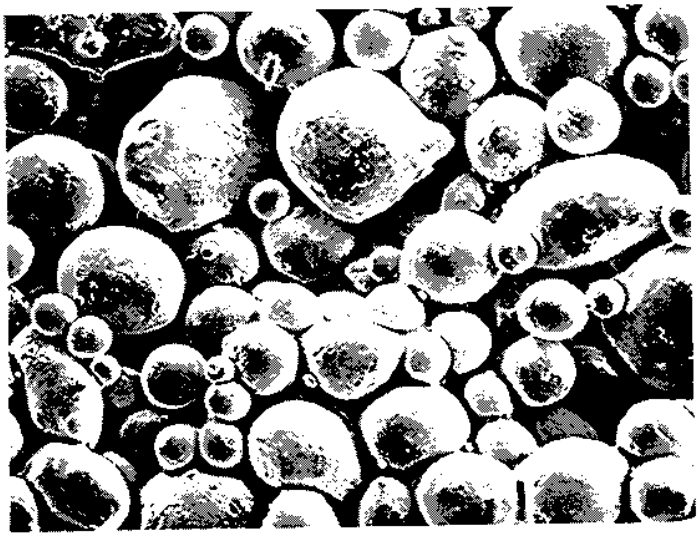

Fig. 3 SEM micrograph of -35 mesh gas atomized commercially pure titanium powder 100x

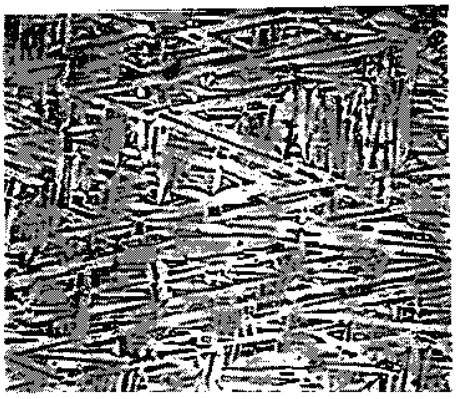

Fig. 4 Microstructure of a Ti-6Al-4V compact treated to produce a broken-up structure

zation techniques have been evaluated, but the plasma rotating electrode process (PREP) is the common commercial method. In the PREP process, a machined bar of the desired composition is used as an electrode which is rotated inside a controlled atmosphere chamber. A plasma torch placed opposite the rotating bar is focused on the bar end causing localized melting at the surface. Molten droplets are spun off the end of the rotating bar by the induced centrifugal force. The droplets solidify in flight producing spherical powder particles which are collected and withdrawn from the chamber. Because the liquid metal makes no contact with any refractory or other material, this process produces powder with no contamination.

Gas atomization (GA) is a non-centrifugal rapid solidification process which has been commercial since 1989. Raw material is melted in a vacuum/inert gas induction furnace with a "cold wall" water cooled copper crucible. As the charge is melted, a thin solid shell of the melt forms at the crucible wall. This shell is called a skull and the process is referred to as skull melting. It is refractory-free because the molten metal comes into contact with no material other than the skull of the alloy itself, thereby eliminating contamination. The molten metal flows through a metal nozzle which directs the liquid into a ring of high pressure jets of inert gas. The gas breaks up the metal stream into tiny spherical particles which solidify in flight, producing powder.

Gas atomization can produce a limitless range of compositions. Its versatility comes from the fact that no specially prepared prefabricated starting stock is required. Instead, raw materials are charged into the crucible to formulate the desired composition. Once the charge melts, it can be held in the molten state for a sufficient period of time to ensure complete homogenization. When induction heating is used for melting, the vigorous stirring action produced by the electromagnetic field also aids in homogenization. Direct atomization from the melt allows for production of highly alloyed compositions, even metastable solid solutions, and minimizes the possibility of segregation in the as-atomized powder. It should also mean lower raw material costs. The stock for PREP powder must be melted into an ingot, worked down to an appropriate size rod, and than machined to tight tolerances. Making powder directly from the melt saves all of these steps.

In contrast to GA, other RS atomization processes which involve localized melting of prealloyed bar stock are limited by the range of alloys available as conventionally produced starting stock. Some complex alloys, unavailable as barstock, may be cast into electrodes. Depending on the composition, there is a risk of thermal shock destroying the cast bar when the localized heat source impinges the electrode. Also, because localized melting methods liquify such a small portion of the metal at any time, compositional variations in the starting stock may directly translate into compositional variations in the resultant powder.

Other Methods of Powder Production. In addition to the HDH and RS methods, several other methods for PA powder production have been developed but none are currently in commercial use. These include:

- *The Hurd Shaker Process* in which the appropriate metal chlorides are heated together with sodium, and then vigorously shaken to exothermically react forming alloy powder and sodium chloride.
- *The ALTi-OXY Process* in which a low melting mixture of titanium dissolved in molten zinc is atomized by any of various methods, and the powder product is subsequently sublimed to remove the zinc.
- The *Goldschmidt Calciothermic Reduction* in which a blend of various metal oxides is essentially reduced with calcium, although several intermediate steps are required to produce the correct soichiometry in the final powder.

Consolidation

Titanium powders can be consolidated into preforms which will undergo further processing to mill products, or into near net shapes (NNS).

Blended elemental (BE) P/M technology involves the blending of titanium powder with a master alloy powder in proper proportions to yield the alloy composition. The blend is typically compacted by mechanical die pressing or cold isostatic pressing. Direct powder rolling and rapid omnidirectional compaction (ROC) may also be used. BE compacts typically have a green density of 85%. Vacuum sintering achieves densities >95%. Careful control of particle size and size distribution can produce compacts that are 99% dense. Porosity is a result of sodium chloride residues in the sponge from the reduction (Kroll) process. Essentially 100% density, and properties equivalent to that of wrought material, can be obtained by using low-Cl sponge.

Sintering is performed at temperatures in the range of 1150 to 1315 °C (2100 to 2400 °F) in a vacuum to prevent gas contamination that can severely degrade compact properties. The high sintering temperature is needed to provide particle bonding and to homogenize the chemistry. It is well above the β transus of all common titanium alloys. The resulting microstructure is much finer than ingot material treated at the same temperature.

Post-sinter hot isostatic pressing (HIP) may be used to further densify the P/M parts. After such treatments BE P/M conpacts are >99% dense and are used as near-net shape in many applications, or are used as preforms for further hotworking to forgings, bar, plate, sheet, or foil. Although not in commercial use, other hot pressing methods have been developed which can be used as an alternative to HIP achieving pseudo-hydrostatic pressure by forge pressing with sophisticated tooling configurations. These include VHP: vacuum hot pressing, ROC: rapid omnidirectional compaction, FDC: fluid die compaction, and the Ceracon process.

Prealloyed (PA) P/M Technology. PA powders produced by HDH have an irregular shape which allows them to be processed by the cold pressing and sintering techniques mentioned above. PA powders produced by RS are spherical and therfore will not cold compact. Hot isostatic pressing is the primary compaction method used for the consolidation of spherical powders. Powder compacts may also be consolidated by direct extrusion or other hot pressing methods developed as an alternative to HIP.

Compaction by hot pressing is typically carried out at a temperature below the beta transus to control microstructure and minimize reaction with the can. Powder cleanliness is one of the main factors governing the quality of PA compacts, because even a low level of contamination with foreign particles may lead to a substantial loss of inherent properties such as fatigue strength.

Postcompaction Treatments. P/M alloys that are subjected to postcompaction working (or to lower-temperature consolidation) may display improved tensile and fatigue strengths as a result of microstructure refinement. However, in most cases, process economics do not allow subsequent working because it nullifies the objectives of a true net-shape technology. Therefore, postcompaction methods leading to microstructure refinement without the use of working are more cost effective. Two approaches that meet this requirement are heat treatment and thermochemical processing.

Heat Treatment. Response to heat treatments after compaction depends on the particular alloy and the P/M processing method. In the case of BE Ti-6Al-4V, for example, the only successfully used heat treatment has been the broken-up

structure (BUS) treatment in which a β quench is followed by 850 °C (1560 °F) long-term annealing. After such treatment, the microstructure of the alloy is showing broken-up α phase in a matrix of β (Fig. 4). This microstructure provides a significant improvement in both tensile and fatigue strengths.

The BUS method is an improvement over standard heat treatments, which typically provide higher tensile properties but no increase in fatigue strength properties. Strengthening by standard heat treatment (such as solution treating and aging) may also degrade the toughness of BE compacts to unacceptable levels. This limitation is possibly due to the porosity from chloride impurities. In contrast, PA compacts may have a better response to solution treatment and aging. Prealloyed compacts of Ti-6Al-4V, for example, have responded well to solution treatment and aging (R.E. Peebles and C.A. Kelto, Investigation of Methods for the Production of High Quality, Low Cost Titanium Alloy Powders, *Powder Metallurgy of Titanium Alloys*, F.H. Froes and J.E. Smugeresky, Ed., The Metallurgical Society of AIME, 1980, p 47-58).

Hydrogenation. The use of hydrogen as a temporary alloying element is used for both postcompaction and precompaction treatment. Thermochemical treatment with hydrogen after compaction refines the microstructure. This refinement provides slightly higher strength levels. Also, it has been demonstrated that starting with powder in the hydrogenated condition provides advantages in processing (lower compaction pressure and/or temperature) and in control of the microstructure for compaction by both hot pressing and hot isostatic pressing. During vacuum hot pressing, the hydrogen is removed during the compaction cycle. In the case of hot isostatic pressing, the hydrogen is retained in the titanium during compaction. After both processes, a final vacuum anneal is required to bring the hydrogen content below specification levels (generally <125 ppm).

Applications

Porous BE Products. On the low end of the density scale, commercially pure (CP) titanium filters are produced for electrochemical and other corrosion-resistant applications. Titanium porous materials are available in sheet and disk form, with controlled porosity ranging from 3 to 40 μm and density ranging from 25 to 75% of theoretical. Sheet is available in widths ranging from 30 to 90 cm (1 to 3 ft). Porous titanium tube also is available in lengths up to 60 cm (2 ft) and with diameters up to 12.7 cm (5 in.).

Compacted BE Products. For more demanding applications, BE components with densities from 98% to close to 100% are produced by cold pressing or CIP (cold isostatic pressing), and sintering followed by HIP (hot isostatic pressing). Very complex shapes such as impellers, are produced by cold isostatic pressing with elastomeric molds. Dimensional tolerances appear to be about ±0.02 mm/mm in length; length is limited by the processing equipment. Part diameter is limited to 600 mm (24 in.) for the CIP method.

Larger parts are produced by press consolidation. Intricate shape capabilities of this method do not approach those of the CIP method, but press consolidation methods appear capable of 0.01 mm/mm in length. Press consolidation is not limited by size; presses up to 45,000 metric tons (50,000 tons) are available in the United States. Consequently, a part of almost 13,000 cm^2 (2,000 $in.^2$) can be produced.

PA powders are used to produce a variety of fully dense products ranging from simple mill products and forging preforms, to near-net shapes for medical aerospace applications. The properties of these products are comparable to wrought products and exceed those of cast products. PA powders are also used in plasma spray deposition to produce sheet, foil, and metal matrix composites.

Currently, P/M shapes from spherical PA powders are produced by the metal can method. The ceramic mold method and the fluid die process have also been developed but neither is used commercially. Both the metal can and ceramic mold processes use hot isostatic pressing consolidation, while the fluid die process is adaptable to either hot isostatic pressing or hot press compaction. The largest HIP chamber available to date is 1350 mm (54 in.) in diameter and 3000 mm (120 in.) in height.

Metal Can Method. The metal can is shaped to the desired configuration by state-of-the-art sheet metal methods, such as brake bending, press forming, spinning, or superplastic forming. Carbon steel is the best suited container material, because it reacts minimally with titanium, forming titanium carbide, which inhibits further reaction. Fairly complex shapes can be produced by this technique.

Ceramic Mold Method. The Crucible ceramic mold process utilizes a ceramic mold produced by the same process used by the investment casting industry. However, the mold materials have been modified to minimize reaction with titanium and to maximize shape capability. Hot isostatic pressing at 815 °C to 1240 °C (1500 °F to 2200 °F) and 103 MPa (15,000 psi) is used to consolidate the powder to full density and form the desired shape. Generally, titanium alloys are consolidated below the beta transus temperature. A prototype impeller made out of titanium aluminide powder has been produced by this method.

The fluid-die process (the ROC process) practiced by Kelsey-Hayes is an outgrowth of earlier work on glass containers. Dies are made from carbon steels, copper alloys, or ceramic materials and are of sufficient mass and dimension to behave as a viscous liquid under pressure at temperature. Dies are comprised of two halves, with inserts where necessary to simplify manufacture. The two halves are then welded together to form an hermetic seal. Powder loading and consolidation follow. The fluid die process combines the ruggedness and fabricability of metal with the flow characteristics of glass to generate a replicating container capable of producing extremely complex shapes. Like the ceramic mold process, a number of shapemaking steps are required to reach the final part configuration. Like the ceramic-mold method, the fluid-die process is capable of generating more complex and intricate shapes than the metal can method.

P/M Property Data CermeTi® Property Data

CermeTi is a trademark registered at the U.S. Office of Patents and Trademarks and is the subject of U.S. and foreign patents assigned to Dynamet Technology, Inc.

CermeTi metal matrix composites are particulate-reinforced titanium metal matrix composites developed by Dynamet

Technology, Inc. This family of materials includes a choice of several ceramic or intermetallic additions (e.g., TiC, TiB_2, or TiAl) at various loading levels in a select titanium alloy matrix such as Ti-6Al-4V. These materials are produced by blended elemental powder metallurgy processing to near-net shape by the proprietary CHIP process and can also be further consolidated/refined by standard forging or extrusion of the P/M preform.

Compared to titanium alloys, CermeTi materials offer improved tensile strength and elastic modulus, both at room temperature and elevated temperature, thereby increasing the use temperature at approximately the same low density. These composites also offer increased hardness and wear resistance over conventional titanium alloys.

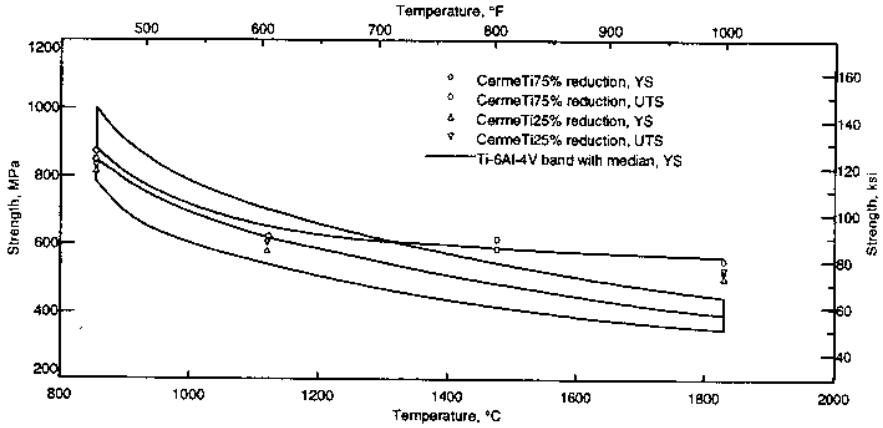

Fig. 1 CermeTi-15: Tensile property comparison vs test temperature

Table 1 CermeTi: Physical properties

Property	Ti-6Al-4V	CermeTi-C-10 10 wt% TiC/Ti-6Al-4V	CermeTi-C-20 20 wt% TiC/Ti-6Al-4V
Density, g/cm^3 (lb/in.^{3})	4.43 (0.160)	4.45 (0.16)	4.52 (0.162)
Tensile strength, MPa (ksi), at:			
RT	896 (130)	999 (145)	1055 (153)
540 °C (1000 °F)	448 (65)	551 (80)	620 (90)
Modulus, GPa (10^6 psi), at:			
RT	113 (16.5)	133 (19.3)	144 (21)
540 °C (1000 °F)	89 (13)	105 (15.3)	110 (16)
Fatigue limit (10^6 cycles), MPa (ksi)	517 (75)	275 (40)	...
Fracture toughness, MPa$\sqrt{m}$ (ksi $\sqrt{in.}$)	55 (50)	44 (40)	32 (29)
Coefficient of linear thermal expansion (RT to 540 °C, or 1000 °F), ppm/°C	8.5	8.1	8.0
Hardness, HRC	34	40	44

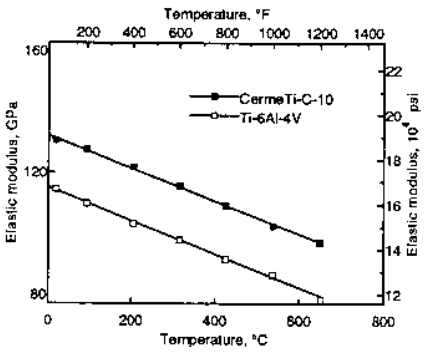

Fig 2 CermeTi with 15% wt% TiC compared to Ti-64. CermeTi-C-10: Dynamic (elastic) modulus vs temperature

Table 2 CermeTi-C: Mechanical property comparison

Material	Density g/cm^3	Density lb/in.^{3}	Yield strength MPa	Yield strength ksi	RT modulus GPa	RT modulus 10^6 psi	Fracture toughness MPa$\sqrt{m}$	Fracture toughness ksi$\sqrt{in.}$
6061-T6 aluminum	2.71	0.098	275	40	68.9	10.0	...	...
2014-T6 aluminum	2.79	0.101	413	60	72.4	10.5	30	28
4340 steel	7.75	0.280	827-1379	120-200	199	29.0	55-110	50-100
25 vol% SiC/6061-T6 Al	2.90	0.105	413	60	113.7	16.5	...	...
Ti-6Al-4V	4.43	0.160	896-1034	130-150	113.7	16.5	55-110	50-100
CermeTi-C-10	4.45	0.161	965	140	133.0	19.3	44	40
CermeTi-C-20	4.48	0.162	1034	150	144.8	21.0	32	29

Ti-1Al-8V-5Fe P/M Property Data

Tensile properties of various product forms of Ti-1Al-8V-5Fe

Room temperature tensile properties of various heat treatments of I/M and P/M Ti-1Al-8V-5Fe. Higher oxygen contents increase strength. Boron additions resulted in zero ductility.

Raw material	Processing	Heat treatment	UTS ksi	TYS ksi	Elongation, %	RA, %
I/M 4 in RCS	Forged	1450 °F/0.5 h/WQ + 950 °F/2 h	213.2	210.7	1.7	5.0
High O_2 (0.34 O wt%)		1450 °F/0.5h/WQ + 1000 °F/2 h	200.5	198.5	2.5	5.0
		1475 °F/0.5 h/WQ + 1050 °F/8 h	186.1	186.1	2.0	6.0
I/M 4 in RCS	Forged	1450 °F/0.5 h/WQ + 1000 °F/6 h	177.4	174.6	1.8	8.5
Low O_2 (0.15 O wt%)		1475 °F/0.5 h/WQ + 1050 °F/10 h	156.6	152.5	5.0	13.1
ALTi P/M	Extruded, 1200 °F, 10:1	1450 °F/0.5 h/WQ + 950 °F/2 h	204.7	194.5	3.0	7.1
		1450 °F/0.25 h/WQ + 925 °F/4 h	200.5	194.3	4.2	...
		1450 °F/8 h/WQ + 925 °F/4 h	198.3	191.0	8.0	15.1
GA PM, Lot 1	Extruded, 1500 °F, 6:1	1425 °F/1 h/WQ + 925 °F/4h	220.0	212.4	7.8	19.5
(0.36 O wt%)	Extruded, 1200 °F, 10:1	1425 °F/0.5/WQ + 925 °F/4 h	225.0	221.0	2.0	4.0
	HIP, 1450 °F/15 ksi/6 h	1425 °F/1 h/WQ + 925 °F/4 h	217.9	207.7	7.0	15.5
GA PM, Lot 2	Extruded, 1500 °F, 6:1	1425 °F/1 h/WQ + 925 °F/4 h	266.5	251.4	...	2.0
(1% B)	Extruded, 1200 °F, 10:1	1425 °F/0.5 h/WQ + 925 °F/4 h	166.3	...	...	...
GA PM, Lot 3	HIP, 1500 °F/15 ksi/6 h	1425 °F/1 h/WQ + 925 °F/4 h	223.6	215.4	4.0	3.2
(0.362 O wt%)		1450 °F/1 h/WQ + 950 °F/4 h	223.8	216.0	4.0	5.7
GA PM, Lot 4	HIP, 1500 °F/15 ksi/6 h	1425 °F/1 h/WQ + 925 °F/4 h	165.3	...	...	...
(.25% B)		1450 °F/1 h/WQ + 950 °F/4 h	114.7	...	...	...

R.R. Boyer, E.R. Barta, Boeing Co., C.F. Yolton, Crucible Materials Corporation, Crucible Research Division, D. Eylon, University of Dayton, "P/M of High Strength Titanium Alloys," P/M in Aerospace and Defense Technologies, Vol 1, p 99-115

Tensile properties of gas atomized P/M Ti-1Al-8V-5Fe

Room temperature tensile properties of GA Ti-1Al-8V-5Fe powder with 0.36 oxygen content after various treatments. Transus: 850 °C (1565 °F)

Consolidations/ conditions	Heat treatment	UTS ksi	TYS (0.2%) ksi	Elongation, %	RA, %
HIP 1500 °F	1250 °F 1 hr AC	163	153	15.6	37.8
HIP 1500 °F	1425 °F 1 h WQ + 925 °F 4 h	219	208	4.7	14.4
HIP 1600 °F	1250 °F 1 h AC	162	152	10.9	19.9
HIP 1600 °F	1425 °F 1 h WQ + 925 °F 4 h	217	205	1.6	7.5
Extrude 1500 °F 6:1	1250 °F 1 h AC	154	142	21.9	48.3
Extrude 1500 °F 6:1	1425 °F 1 h WQ + 900 °F 4 h	222	212	9.4	15.0
Extrude 1500 °F 6:1	1425 °F 1 h WQ + 925 °F 4 h	220	212	7.8	19.5
Extrude 1500 °F 6:1	1500 °F 1 h WQ + 925 °F 4 h	227	219	6.3	7.0

C.F. Yolton, Crucible Materials Corporation, Crucible Research Division, unpublished data, 1992

Tensile properties of gas atomized P/M Ti-1Al-8V-5Fe

Room temperature tensile properties of GA Ti-1Al-5Fe powder with 0.14 oxygen content after various heat treatments. Transus: 750 °C (1390 °F)

HIP temperature		Heat	Tensile strength		Yield strength 0.2% offset		Elong-ation,	RA,
°C	°F	treatment(a)	MPa	(ksi)	MPa	(ksi)	%	%
725	1340	As-HIP	1063	(154.3)	1037	(150.5)	6.3	29.1
		A	1007	(146.2)	960	(139.4)	17.2	39.8
		B	1407	(204.2)	1397	(188.2)	6.3	8.0
		C	1217	(176.7)	1160	(168.4)	12.5	27.8
780	1440	As-HIP	1135	(164.7)	1075	(154.6)	9.4	19.9
		A	1015	(147.4)	960	(139.4)	16.4	37.2
		B	1414	(205.2)	1315	(190.8)	4.7	9.9
		C	1175	(170.5)	1111	(161.3)	10.9	24.3

(a) Heat treatment: A - 675 °C (1250 °F) 1 h, AC, B - 675 °C (1250 °F) 1 h, WQ/440 °C (825 °F) 4 h, C - 675 °C (1250 °F) 1 h, WQ/495 °C (925 °F) 4 h. Source: C.F. Yolton, J.H. Moll, Crucible Materials Corporation, Crucible Research Division, "Evaluation of a High Strength Rapidly Solidified Beta Titanium Alloy," Progress in Powder Metallurgy, 1987, Vol 43, MPIF 1987, p 49-63

Typical P/M Property Data

Tensile properties of P/M beta Ti alloys

Each of the alloys was HIPed at 15 °F-25 °F above the transus or at 50 °F-60 °F below the transus. The HIP cycle was 12 h at 15 ksi.

Alloy	Heat treatment	Tensile strength (ksi)	Yield strength 0.2% offset (ksi)	Elongation (%)	Reduction of area (%)	Modulus 10^6 psi
Ti-10-2-3	As HIP 1400 °F	128.5	119.2	22.0	60.4	13.4
	1410 °F 1 h WQ + 900 °F 8 h	208.8	188.0	3.0	2.6	15.0
		207.3	186.7	4.0	6.2	13.5
	1375 °F 1 h WQ + 950 °F 8 h	173.5	158.1	10.0	29.6	15.0
	1410 °F 1 h WQ + 950 °F 8 h	186.7	171.7	8.0	15.9	14.9
		187.2	172.0	9.0	22.9	14.4
	1440 °F 1 h WQ + 950 °F 8 h	194.6	177.5	...	...	14.7
	As HIP 1475 °F	133.0	118.4	21.0	58.0	13.6
	1410 °F 1 h WQ + 900 °F 8 h	212.2	191.5	3.0	6.2	14.4
		211.0	192.6	2.0	6.2	14.4
	1375 °F 1 h WQ + 950 °F 8 h	172.5	155.1	11.0	31.8	14.7
	1410 °F 1 h WQ + 950 °F 8 h	194.4	180.1	6.0	16.4	15.0
		194.0	176.4	5.0	16.2	15.3
	1440 °F 1 h WQ + 950 °F 8 h	197.4	180.7	5.0	7.3	14.7
Beta III	As HIP 1325 °F	154.9	143.4	14.0	39.8	12.0
	1325 °F 5 h WQ + 900 °F 8 h	199.2	185.9	4.0	6.5	14.0
		200.3	186.8	6.0	9.2	14.0
	1275 °F 5 h WQ + 950 °F 8 h	176.7	165.0	9.0	22.7	14.1
	1325 °F 5 h WQ + 950 °F 8 h	185.6	174.3	6.0	9.0	14.1
	1375 °F 5 h WQ + 950 °F 8 h	191.7	179.8	7.0	12.9	14.0
	As HIP 1400 °F	152.3	140.1	12.0	32.5	10.8
	1325 °F 5 h WQ + 900 °F 8 h	201.3	186.4	3.0	4.4	13.7
		195.3	181.9	2.0	5.6	13.6
	1275 °F 5 h WQ + 950 °F 8 h	171.5	155.4	8.0	16.7	14.7
	1325 °F 5 h WQ + 950 °F 8 h	184.4	171.6	6.0	11.6	13.8
		184.3	170.9	6.0	12.1	13.9
	1375 °F 5 h WQ + 950 °F 8 h	181.7	166.4	8.0	18.3	14.0
Alloy 253	As HIP 1400 °F	127.0	90.5	20.0	45.0	12.0
	1425 °F 1 h WQ + 900 °F 8 h	209.6	196.7	2.0	4.4	14.7
		209.3	195.6	2.0	4.0	14.9
	1350 °F 1 h WQ + 950 °F 8 h	184.5	172.6	7.0	12.5	14.9
	1425 °F 1 h WQ + 950 °F 8 h	196.5	185.1	3.0	5.8	14.6
		197.8	186.1	3.0	6.4	14.8
	1450 °F 1 h WQ + 950 °F 8 h	198.4	185.7	4.0	6.0	15.8
	As HIP 1475 °F	127.2	121.1	15.0	41.2	11.3
	1425 °F 1 h WQ + 900 °F 8 h	195.0	181.2	2.0	4.6	14.3
		194.2	180.8	2.0	4.6	14.4
	1350 °F 1 h WQ + 950 °F 8 h	182.1	169.6	3.0	4.4	15.8
	1425 °F 1 h WQ + 950 °F 8 h	184.2	170.0	3.0	6.2	14.6
		185.6	172.2	4.0	6.8	14.6
	1450 °F 1 h WQ + 950 °F 8 h	185.6	171.9	4.0	7.0	15.4

Source: C.F. Yolton, Crucible Materials Corporation, Crucible Research Division, "P/M Beta Titanium Alloys for Landing Gear Applications," Progress in Powder Metallurgy, 1986, Vol 42, MPIF 1986, p 635-653

P/M Ti-6Al-2Sn-4Zr-2Mo-0.1Si

Gas atomized P/M Ti-6Al-2Sn-4Zr-2Mo-0.1Si meets tensile and creep properties for wrought material as specified by AMS 4975.

Test temperature	Room temperature	900 °F
UTS (ksi)	151	104
Yield (ksi)	135	82
% Elongation	19.0	12.0
% R.A.	42.5	20.2

Gas atomized powder, –35 and 80 mesh consolidated by HIP, HIP 1700 °F 14 ksi 4 h, solution treat of 1 h 1765 °F AC, plus age 8 h 1100 °F AC. Source: Unpublished data 1991, C.F. Yolton, Crucible Materials Corporation, Crucible Research Division